CS228: Introduction to Deep Learning

Lecture 1: introduction

Instructor: Zhouxing Shi

Logistics

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AI is everywhere

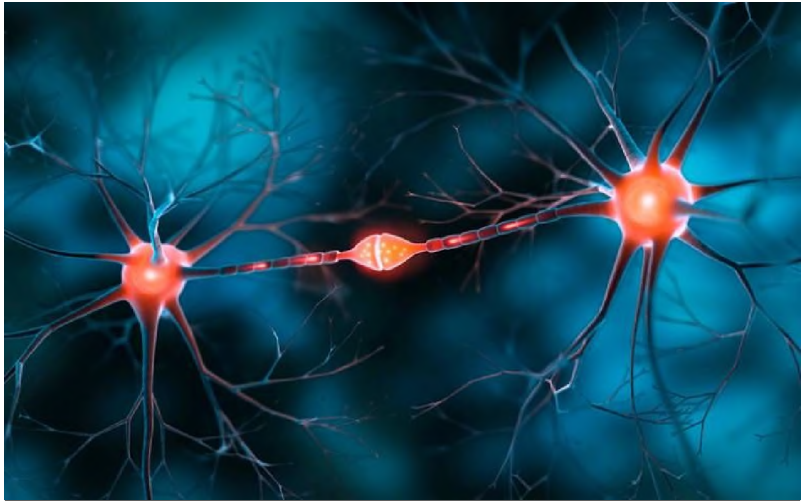
Generative AI

Autonomous systems

Healthcare / scientific discovery

Deep learning behind AI applications

- Most modern AI applications are backed by **deep learning**
- **Neural networks** that are motivated by human brains



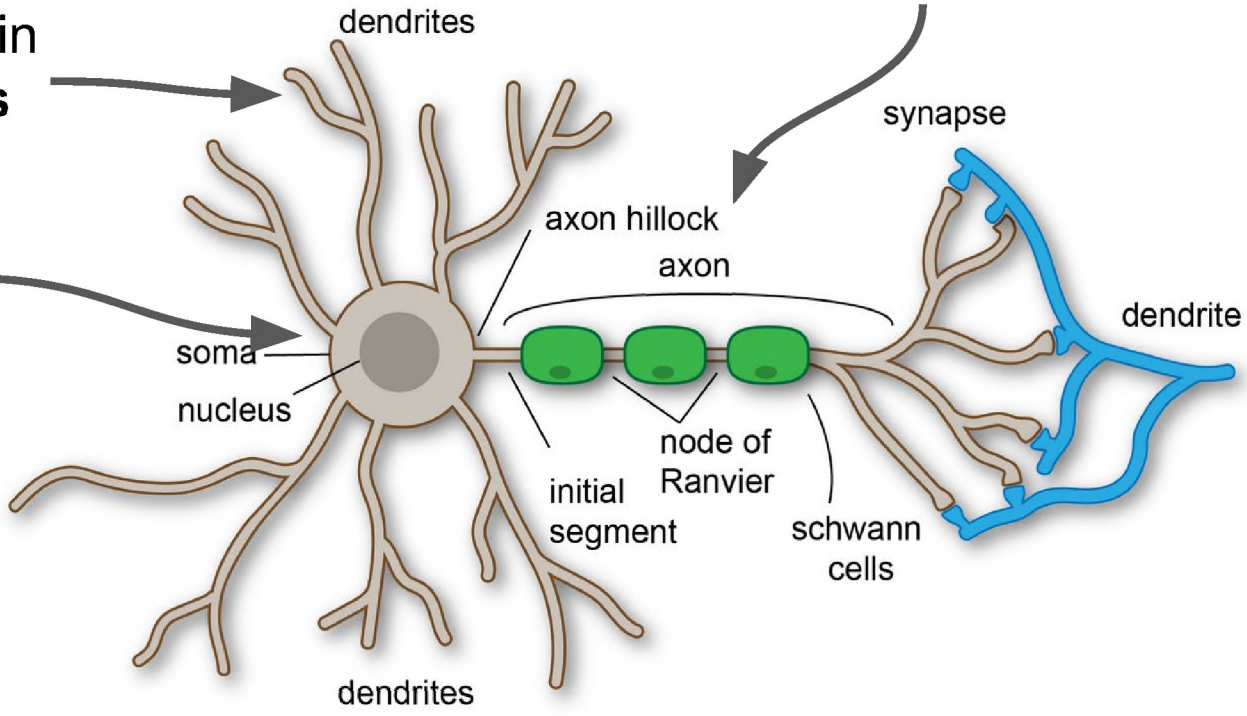
The human brain contains a massive **network of neurons**

Human brain

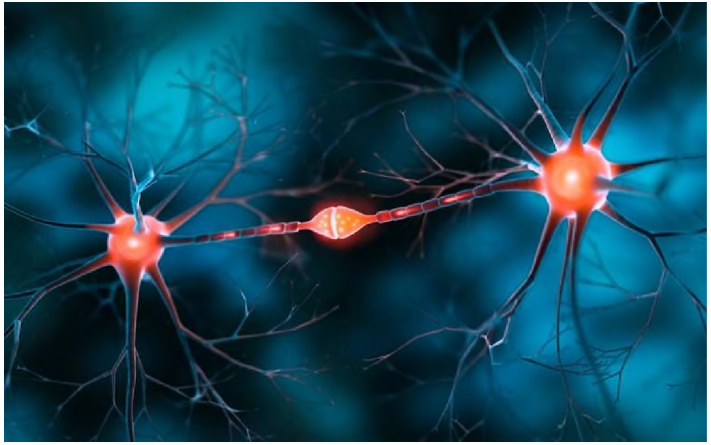
Signal come in via **dendrites**

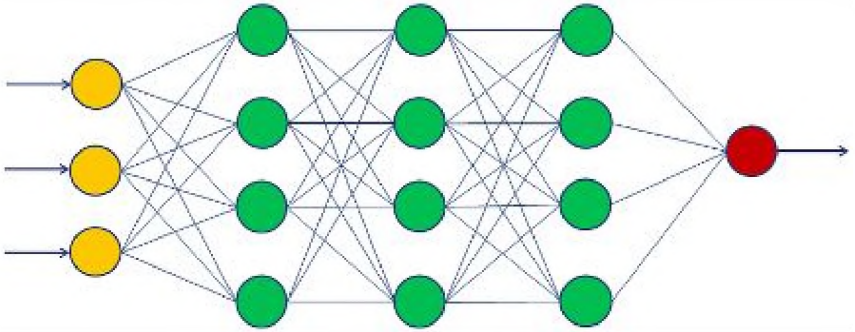
into **soma**

Signal goes out via the **axon**



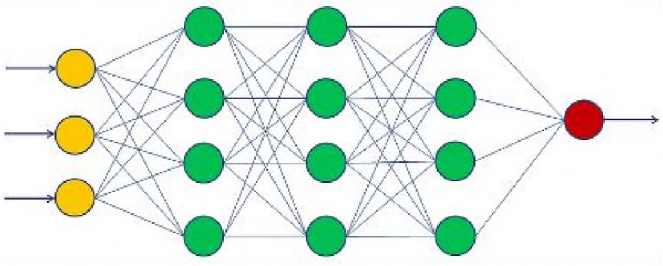
Artificial neural networks



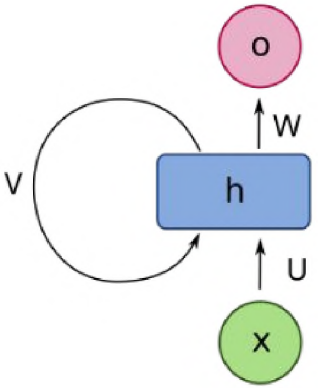


Multi-layer perceptrons (MLP)

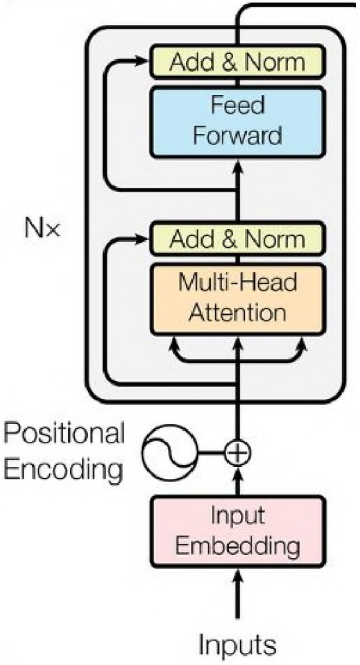
Artificial neural networks



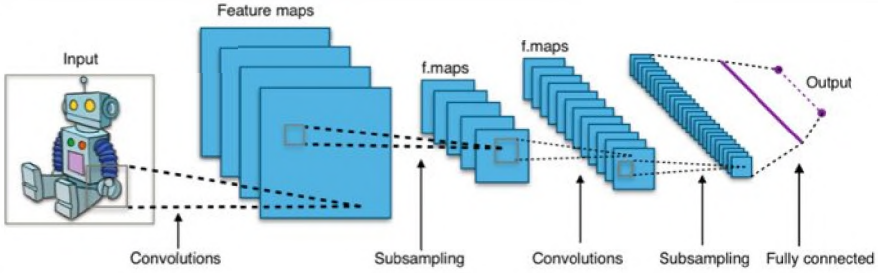
Multi-layer perceptrons (MLP)



Recurrent neural networks



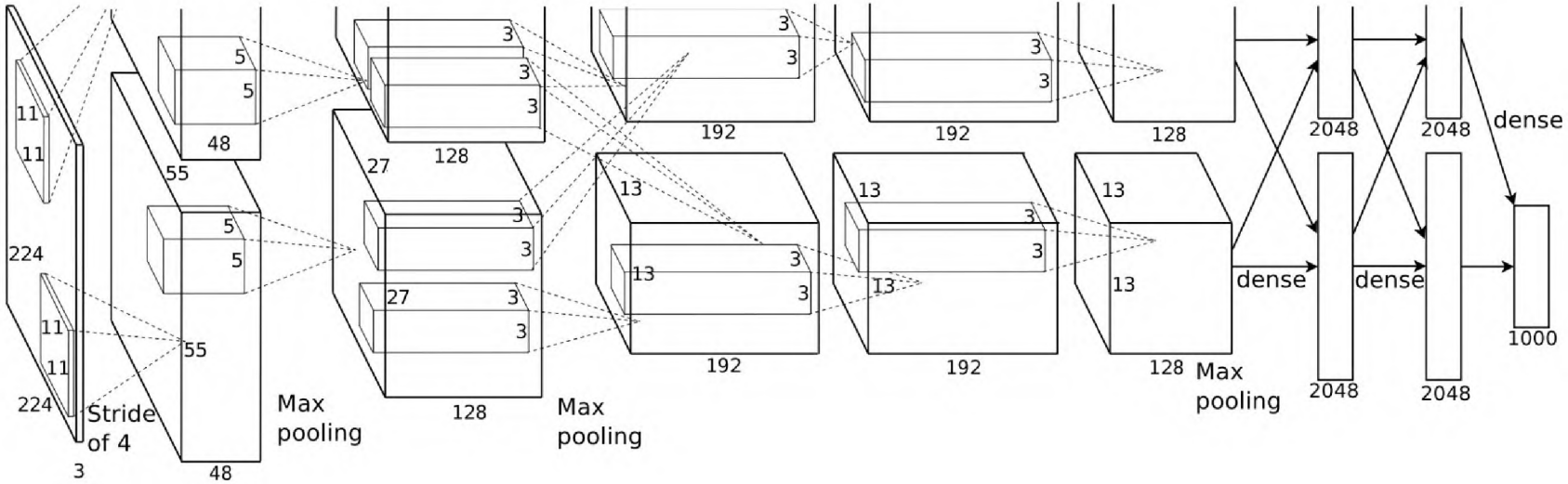
Transformers



Convolutional Neural Networks

Deep neural networks

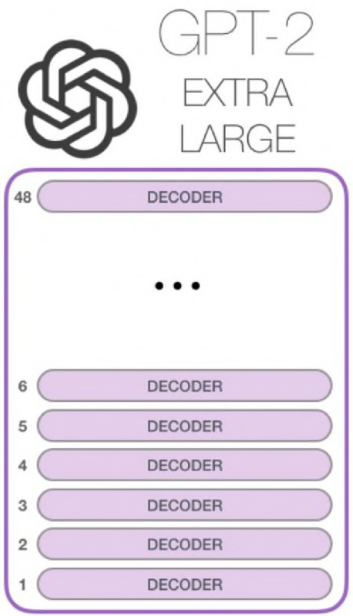
- Deep models for large learning capacity:



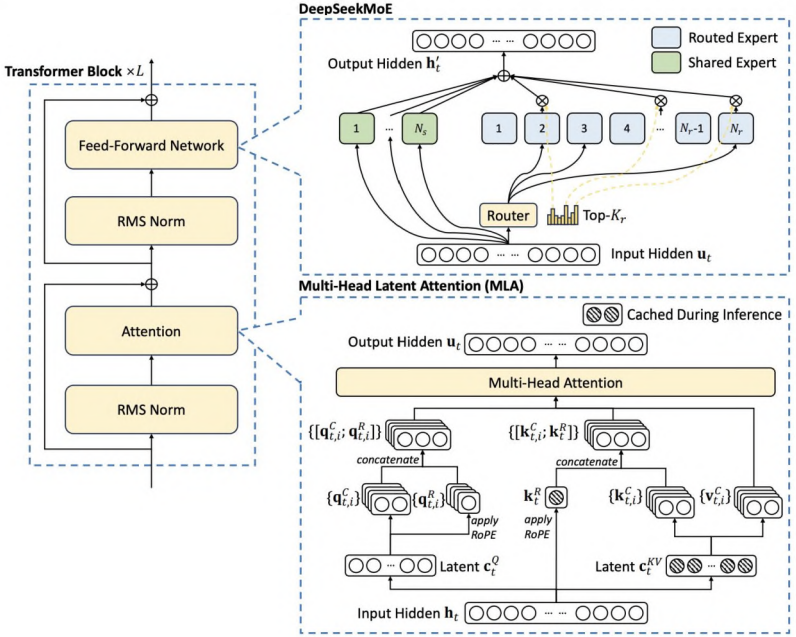
AlexNet in 2012: **60 million** parameters and 650,000 neurons

Deep neural networks

- Modern large foundation models:



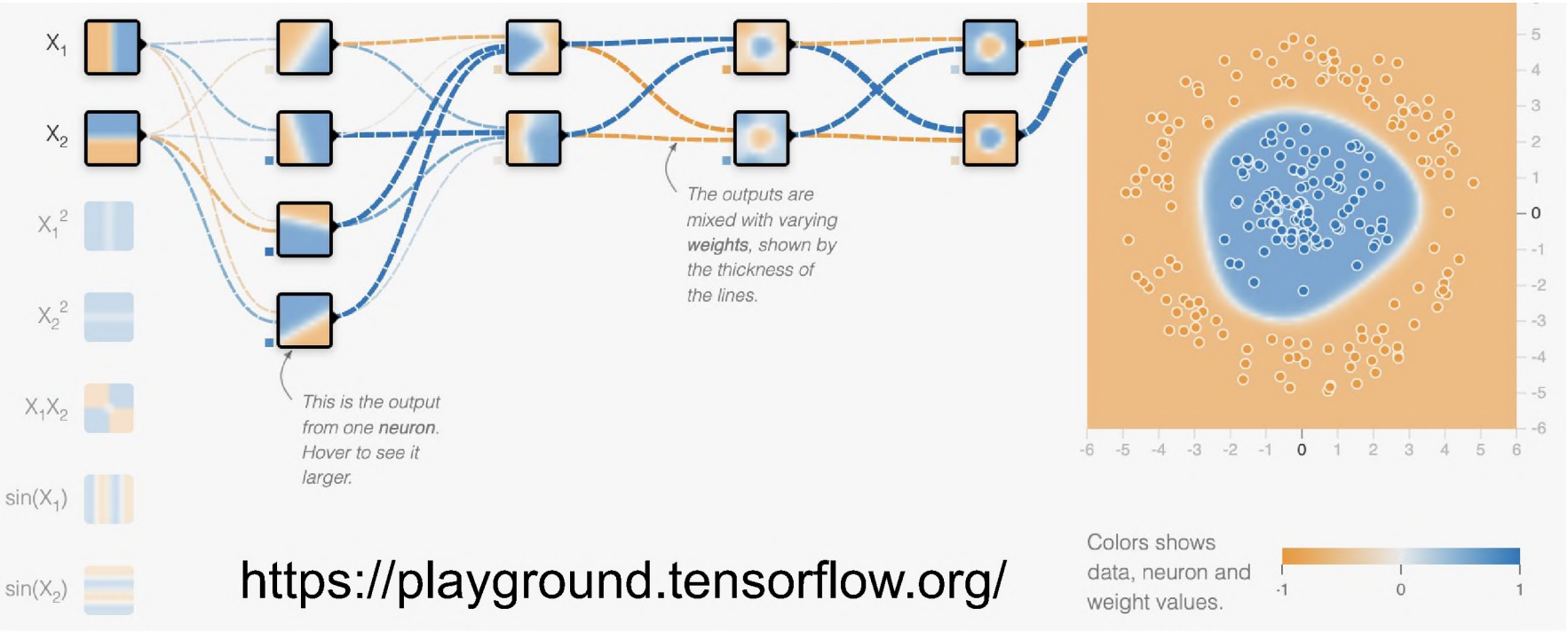
GPT-2 in 2019: **1.5B** parameters



DeepSeek: **236B** in 2024 (V2)
685B in 2025 (V3.2)

Representation learning

- Parameters in models have to be learned from data:



Representation learning

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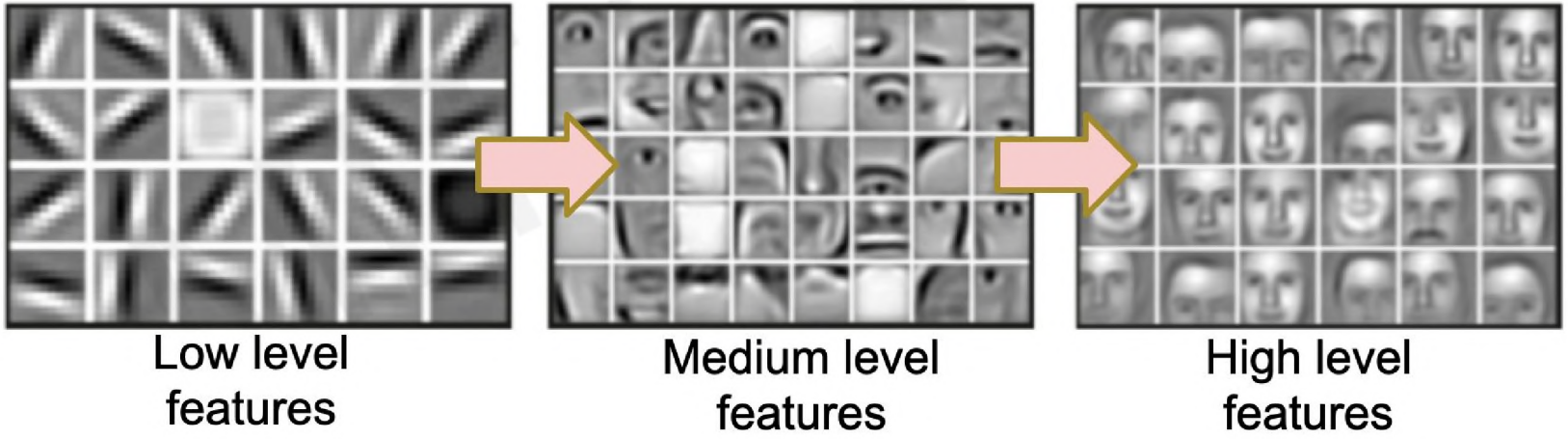
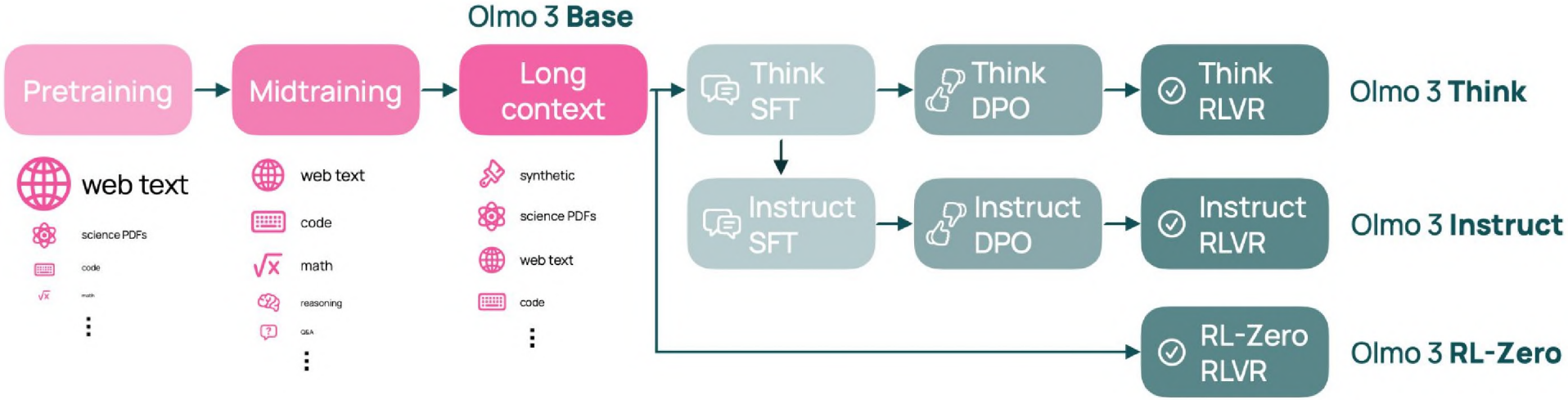


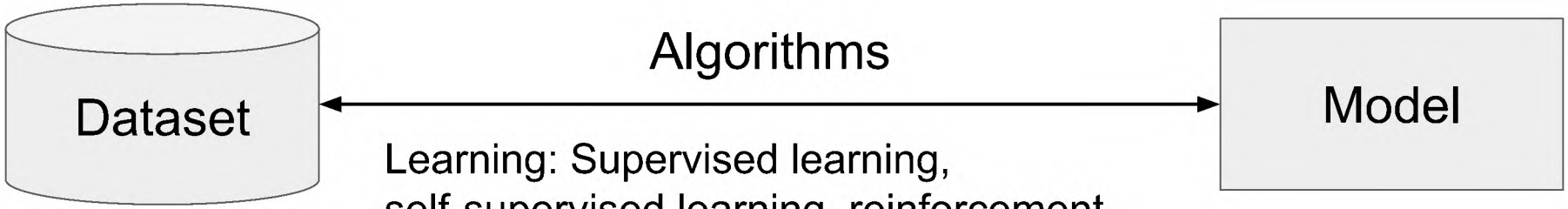
Large-scale training data

- Trillions of training tokens for pre-trained models:



6T **100B** **50B**

Training

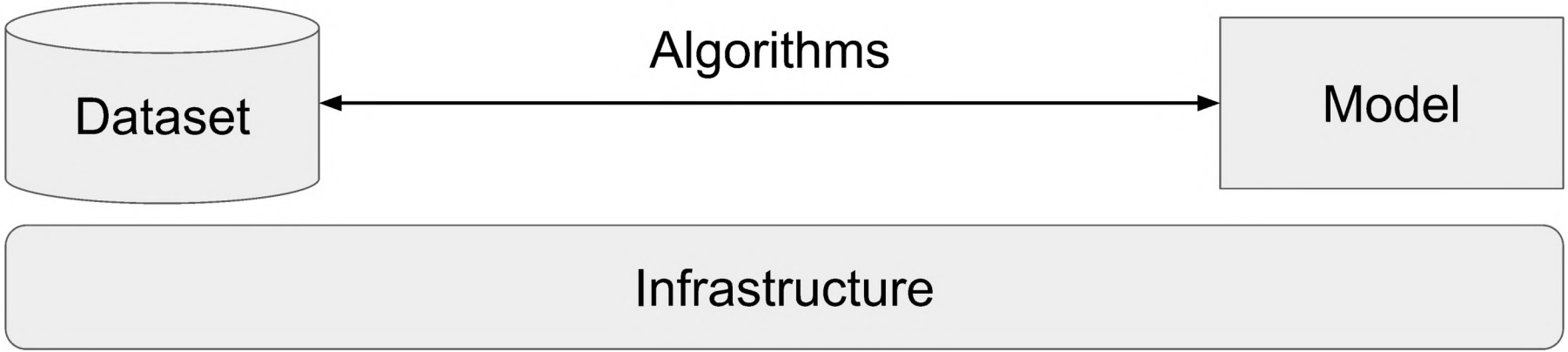


Learning: Supervised learning, self-supervised learning, reinforcement learning, ...

Optimization: Stochastic Gradient Descent (SGD), Adaptive Moment Estimation (Adam), ...

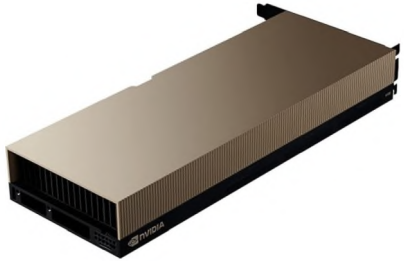
Techniques: regularization, normalization, scheduling, ...

Infrastructure



PyTorch

vLLM



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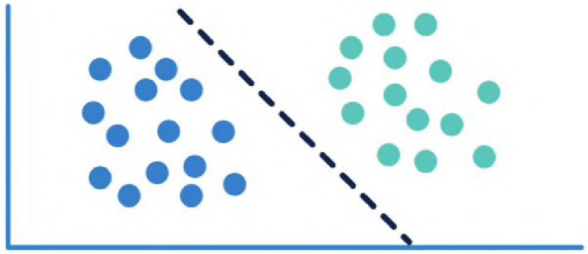
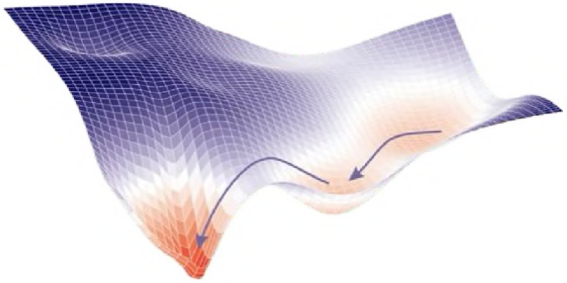
Topics

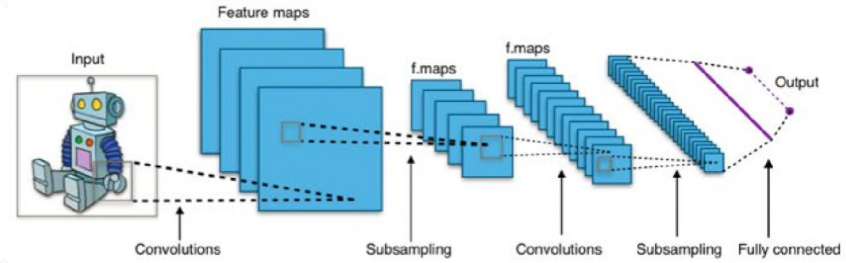
Learning problem



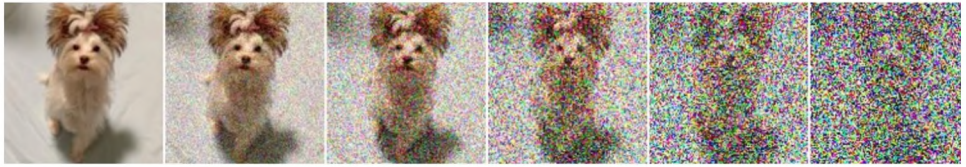
Learning / optimization algorithms



Models architectures



Applications



This course

- Fundamentals of deep learning
  - Learning problem
  - Learning algorithms (backpropagation, optimization, ...)
  - Learning techniques (loss functions, regularization, generalization, ...)
  - Neural network architectures (CNN, RNN, transformers)
- Advanced learning: generative models, self-supervised training, ...
- Beyond learning: efficiency, interpretability, ...

Learning outcomes

- Understand the fundamentals behind deep learning
  - What does deep learning consist of
  - Why are they needed and why do they work
  - How to make them useful in practice
- Build your own deep learning models
- Work on a real problem and address practical challenges

Canvas

- Primary channel for online communication
- Course homepage, syllabus, links to resources: https://elearn.ucr.edu/courses/228263
- Discussion: Canvas -> Discussions
- Announcements, assignments, files

Grading

- Assignment: 30%
- Midterm exams: 30%
- Project: 40%
- Participation bonus: up to 5%

Assignments (30%)

- About three coding assignments
  - To get started with deep learning in PyTorch
    - You should be able to program in Python
  - To understand model architectures and design choices
  - To gain hands-on experience with deep learning

- Should be individually completed
- Discussion with peers is allowed but should be acknowledged.
- You should not use AI to generate solutions
- Late penalty = 3^k (k is the number of days late, rounded up)

Midterm exams (30%)

- Two *async* midterm exams at CSE test center
  - On Week 4 and Week 8
  - 50-min window between Thursday and Friday
- No lecture on those two Fridays

Project (40%)

- A team project to strengthen hands-on experience in deep learning
- Each team should typically have 3 students
- Implementation:
  - AI coding is allowed.
  - Use of open-source code is encouraged but should be documented.
  - New contributions should be demonstrated.
- Deadlines are strict

Project (40%)

- Project types:
  - Engineering-focused: An application of deep learning
    - Design and implement a deep learning pipeline
    - Develop a simple demo
  - Research-focused: A new model / algorithm for a deep learning problem
    - Study existing methods and design a well-motivated new method
    - Comparison with existing approaches (negative results are fine)
    - Analysis of the method and results
  - Research-focused: A novel empirical or theoretical analysis of existing approaches
- Some project directions will be suggested during the class

Project (40%)

- Sign up (due on 4/12, 5%): Form a team and sign up.
- Proposal (due on 4/19, 10%):
  - Choose a task for your project and submit a 1-page writeup.
  - 1) An introduction and the background of the task;
  - 2) An overview of existing methods, including deep learning approaches and/or others (if any);
  - 3) Datasets and metrics.
  - You do not have to propose any new method at this stage.

Project (40%)

- Milestone report (due on 5/6, 15%)
  - A new writeup, updated and extended from your proposal. Add:
  - 1) Initial progress: reproduction of one or two baseline methods (reuse existing code if possible), or an implementation of a straightforward or naive method;
  - 2) Proposal on next steps that propose and implement a new method, conduct a new analysis, or develop a new tool or demo.
- Final presentation (in-class on Week 10, 20%):
- Final report and code submission (due on 6/12, 50%)

Project (40%)

- Midterm project discussion (Friday, 5/8, optional):
  - Individual project discussions between each group and the instructor on Week 6
  - 20-min slots for Zoom meetings
  - In class + extended availability as needed

Tentative schedule

Week	Monday	Wednesday	Friday
1	Introduction	Perceptrons	Perceptrons
2	Backpropagation	Optimization for NNs	Optimization for NNs
3	Training techniques	Convolutional NNs	Convolutional NNs
4	Recurrent NNs	Recurrent NNs	**[Async at test center] Midterm I**
5	Attention	Transformers	Transformers
6	Transformers	Self-supervised learning	**[Zoom slots] Project discussion**
7	Autoencoders	Variational Auto Encoders	Generative Adversarial Networks
8	Diffusion	Diffusion	**[Async at test center] Midterm II**
9	**[No Class]** Memorial Day	Efficiency	Efficiency
10	Interpretability	**Project presentations**	**Project presentations**